

Rihua Van Steenburgh

Data Scientist | Applied AI & Machine Learning

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Summary

Data Scientist focused on applied AI, machine learning, and NLP, with hands-on experience building and evaluating ML and RAG applications. Skilled in Python, SQL, scikit-learn, PyTorch, Transformers, PostgreSQL/pgvector, and FastAPI, with strengths in leakage-safe model evaluation, hybrid retrieval, citation validation, and human-in-the-loop decision support.

Skills

- **Programming & Analysis:** Python, SQL, R, pandas, statistical analysis
- **Data Science & Machine Learning:** scikit-learn, TF-IDF, Linear SVM, PyTorch, Transformers, DistilBERT, text classification
- **Model & Retrieval Evaluation:** Macro F1, model comparison, leakage-safe validation, temporal validation, retrieval evaluation, error/failure analysis
- **Applied AI & RAG:** embeddings, vector search, hybrid retrieval, RRF, PostgreSQL/pgvector, grounded generation, citation validation, explicit abstention handling
- **Data, Cloud & Production:** PostgreSQL, PySpark, Delta Lake, dbt, Databricks, Azure, AWS, FastAPI, Docker, GitHub Actions, Airflow

Projects

- 1. Financial Complaint Auto-Routing with NLP | Python • scikit-learn • TF-IDF • Linear SVM • PyTorch • DistilBERT**
Jun 2026 - Aug 2026 [GitHub](#) | [Case Study](#)
 - Identified and remediated 39.39% normalized-text leakage in the original test split; rebuilt group-aware 2024 development/final-test splits with zero normalized-text overlap.
 - Compared TF-IDF + Linear SVM with a frozen DistilBERT challenger; DistilBERT increased Macro F1 from 0.7671 to 0.7949 on the shared 2024 benchmark, with both frozen models evaluated on a 30,156-row retrospective 2025 cohort.
- 2. CivicLens RAG Operations Copilot | Python • Hybrid Retrieval • PostgreSQL/pgvector • Retrieval Evaluation**
May 2026 - Aug 2026 [GitHub](#) | [Live Demo](#) | [Case Study](#)
 - Built a hybrid RAG retrieval system combining semantic search, PostgreSQL full-text retrieval, and deterministic Reciprocal Rank Fusion (RRF) over curated NYC 311 knowledge, served through FastAPI.
 - Evaluated retrieval and application behavior with citation-validity and abstention checks; hybrid retrieval achieved 83.9% Recall@5 and 92.9% expected-source retrieval across 14 retrieval-eligible questions.
- 3. NYC 311 Service Requests Lakehouse | Azure • Databricks • PySpark • SQL • Delta Lake**
Mar 2026 - May 2026 [GitHub](#) | [Case Study](#)
 - Built an Azure lakehouse pipeline using Azure Data Factory, ADLS Gen2, Databricks, PySpark, SQL, and Delta Lake to prepare Bronze, Silver, and Gold datasets.
 - Created analytics-ready fact tables and Gold marts using reusable cleaning, SQL validation, and star-schema modeling.
- 4. Cloud Flight Fare Pipeline | AWS • Python • SQL • S3 • Redshift • dbt**
Feb 2026 - May 2026 [GitHub](#) | [Case Study](#)
 - Built an AWS data pipeline using Python, S3, Redshift Serverless, SQL, and dbt to ingest and transform flight fare API data.
 - Developed dbt staging models, dimensional marts, and data quality tests to support reproducible downstream analytics.

Education

- **Master of Science (M.S.) in Information Technology, Data Analytics Concentration** - Middle Georgia State University, Expected Dec 2026 | GPA: 4.00
- **Associate of Applied Science (A.A.S.) in Website Design/Development** - Gwinnett Technical College, Dec 2024 | GPA: 3.96
- **Bachelor of Arts (B.A.) in Chinese Ethnic Language and Literature** - Guangxi University for Nationalities, China

Experience

IT Intern - Fulton County | Jan 2024 - Apr 2024

- Standardized IT asset and network inventory records, improving data consistency and audit readiness.
- Reconciled records across multiple systems to support inventory reporting and internal documentation.

Floral Assistant - Kroger | Dec 2022 - Aug 2023

- Reconciled inventory, pricing, and shrink records to improve reporting accuracy and exception tracking.